

Scenario:

A bank conducted direct marketing campaigns (phone calls) to sell term deposits. Your task is to develop a machine learning model to predict whether a client will **subscribe to a term deposit** based on provided customer and campaign attributes.

Dataset:

Use the **Bank Marketing Dataset** Provided in the Learning Management System
Available also from Kaggle via the following URL

URL: <https://www.kaggle.com/datasets/henriqueyamahata/bank-marketing>

Dataset features include:

- Age, job, marital status, education, default, balance, housing loan, personal loan, contact type, day, month, campaign, duration, previous outcome, etc.
- Target variable: **"y" = yes/no (client subscribed to term deposit)**

Tasks:

1. **Prepare training and test datasets.** Explain data cleaning, encoding, normalisation, and splitting.
2. **Select algorithms** (e.g., Logistic Regression, Random Forest, SVM, k-NN).
3. **Train models** and compare performance metrics (accuracy, precision, recall, F1 score, confusion matrix).
4. **Discuss underfitting, overfitting, and balance** in results.
5. **Evaluate and justify** which model performs best and why.
6. **Provide visualisations** (e.g., ROC curve, feature importance).